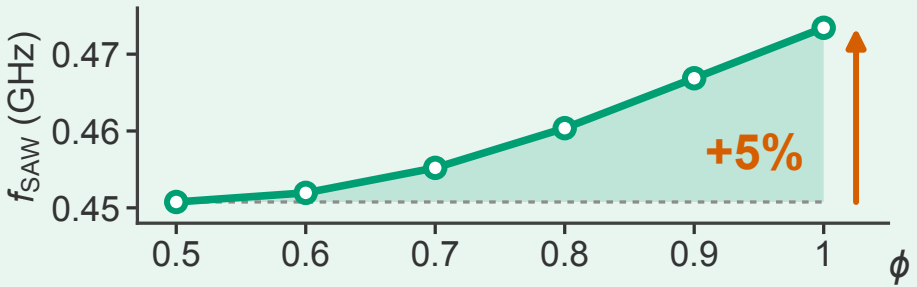
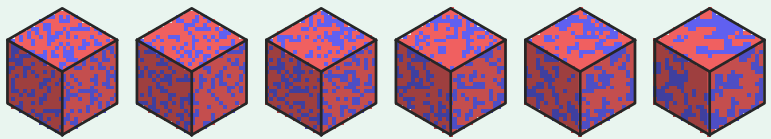
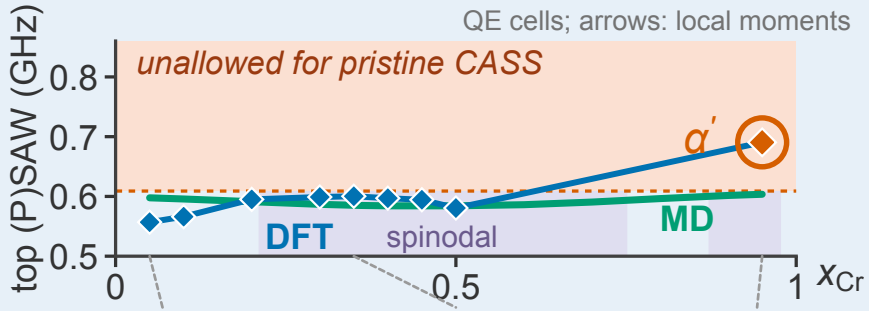
MD Decomposition stiffens the SAW

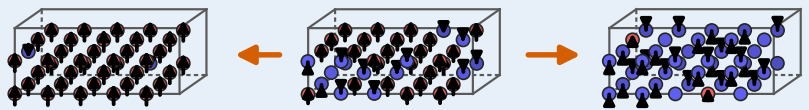




$Fe_{64}Cr_{36}$ (MD): random ⟶ $\alpha + \alpha'$

DFT Stiff α' enters the unallowed range





α	Fe–36Cr	α'
C' = 56 GPa	C' = 68 GPa	C' = **145 GPa**

Decomposition ⟶ stiffer δ-ferrite (C' ↑) ⟶ higher SAW frequency ⟶ non-destructive sign of embrittlement